

Experiments in Spoken Dialogue Systems: From Word Embeddings to Agentic Task-Oriented Dialogue

Anonymous ACL submission

001	Abstract	3.1.2 Training Configuration	029
002			030
003	1 Introduction	3.1.3 Evaluation and Analysis	031
004			032
005	1.1 Word Embeddings	3.2 Dialogue State Tracking with TripPy	033
006			034
007	1.2 Task-Oriented Spoken Dialogue Systems	3.2.1 Architecture, Inputs, and Objective	035
008			036
009	1.3 Report Scope	3.2.2 Training Dynamics	037
010			038
011	2 Experimental Setup	3.2.3 Ablation Study and Analysis	039
012			040
013	2.1 Simple MultiWOZ 2.1	3.3 Dialogue Policy with PPO	041
014			042
015	2.2 ConvLab3 and the Baseline Modular Pipeline	3.3.1 PPO Setup and Policy Variants	043
016			044
017		3.3.2 RL Training Results	045
018	2.3 General Experimental Configuration		046
019		3.3.3 Interaction Quality	047
020	2.4 Implementations and Repositories		048
021		4 Agentic Task-Oriented Dialogue	049
022	3 Results and Analysis		050
023		4.1 Free-Form Web-Enabled LLM	051
024	3.1 Natural Language Understanding with BERTNLU		052
025		4.2 Structured Modular Prompting	053
026			054
027	3.1.1 Task and Model	4.3 Comparison and Deployment Risks	055
028			056

5 Conclusion

References

- Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. 2017. [Enriching word vectors with subword information](#). *Transactions of the Association for Computational Linguistics*, 5:135–146.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Mihail Eric, Rahul Goel, Shachi Paul, Abhishek Sethi, Sanchit Agarwal, Shuyang Gao, Adarsh Kumar, Anuj Goyal, Peter Ku, and Dilek Hakkani-Tur. 2020. [MultiWOZ 2.1: A consolidated multi-domain dialogue dataset with state corrections and state tracking base-lines](#). In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 422–428, Marseille, France. European Language Resources Association.
- Michael Heck, Carel van Niekerk, Nurul Lubis, Christian Geischauser, Hsien-Chin Lin, Marco Moresi, and Milica Gašić. 2020. [TripPy: A triple copy strategy for value independent neural dialog state tracking](#). In *Proceedings of the 21st Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 35–44, 1st virtual meeting. Association for Computational Linguistics.
- Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S. Corrado, and Jeffrey Dean. 2013. [Distributed representations of words and phrases and their compositionality](#). In *Advances in Neural Information Processing Systems 26*, pages 3111–3119. Curran Associates, Inc.
- John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. 2016. [High-dimensional continuous control using generalized advantage estimation](#). In *International Conference on Learning Representations*.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. [Proximal policy optimization algorithms](#). *arXiv preprint arXiv:1707.06347*.
- Qi Zhu, Christian Geischauser, Hsien-Chin Lin, Carel van Niekerk, Baolin Peng, Zheng Zhang, Shutong Feng, Michael Heck, Nurul Lubis, Dazhen Wan, Xiaochen Zhu, Jianfeng Gao, Milica Gašić, and Minlie Huang. 2023. [ConvLab-3: A flexible dialogue system toolkit based on a unified data format](#). In *Proceedings of the 2023 Conference on Empirical*

Methods in Natural Language Processing: System Demonstrations, pages 106–123, Singapore. Association for Computational Linguistics.

A Example Dialogues

A.1 Baseline ConvLab3 Pipeline

A.2 PPO Dialogue Policies

A.3 Free-Form LLM and Boundary Probes

A.4 Structured-Prompt LLM

B Final Structured Prompt